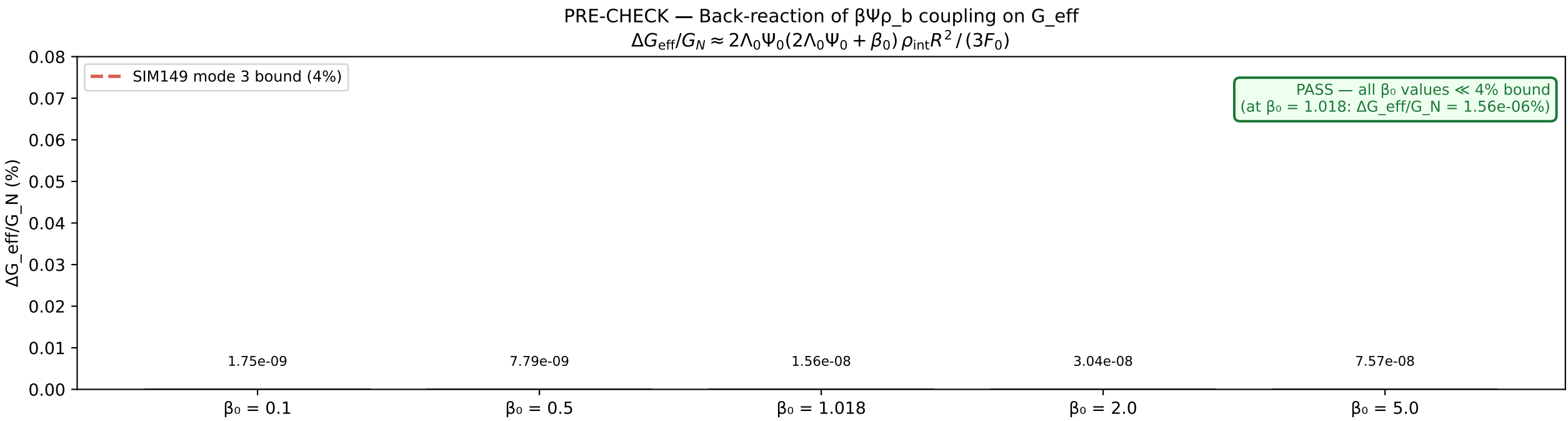




Derivation: Extended Action, Field Equations, Screening

Extended action:

$S = \int d^4x \sqrt{(-g)} [F(\Psi)R - \frac{1}{2}(\partial\Psi)^2 - V(\Psi) + \beta(\rho^b)\Psi\rho^b] + S_{SM}$
 $F(\Psi) = \frac{1}{2} + \Lambda_0\Psi^2, \quad \beta(\rho^b) = \beta_\infty + (\beta_0 - \beta_\infty) \tanh^2(\rho^b/\rho_{sc})$

Modified Klein-Gordon:

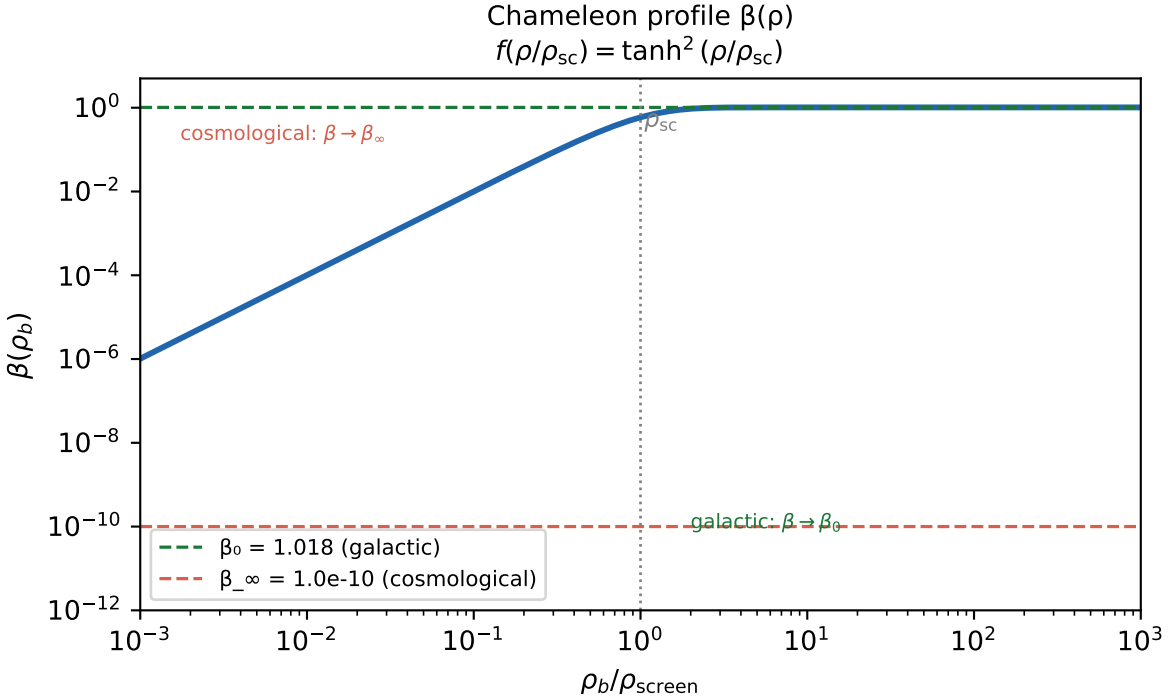
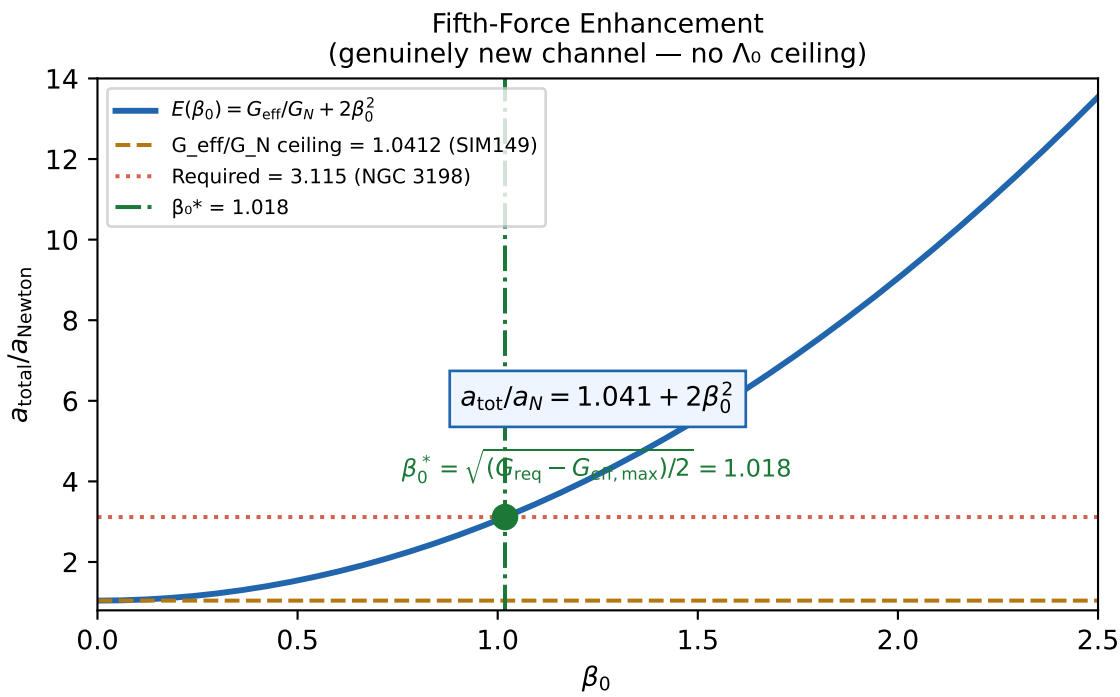
$\square\Psi = V'(\Psi) - 2\Lambda_0\Psi R - \beta(\rho^b)\rho^b$
 $\Rightarrow \nabla^2\Psi = m^2\Psi - 2\Lambda_0\Psi\rho^b - \beta(\rho^b)\rho^b \quad (\text{static, non-relativistic})$

Modified Einstein eqns:

$2F(\Psi)G_{\mu\nu} + [g_{\mu\nu}\square - \nabla_\mu\nabla_\nu](2F) = T_{\mu\nu}^\Psi(\Psi) + T_{\mu\nu}^\Psi(SM) + T_{\mu\nu}^\Psi(\beta)$
 $T_{\mu\nu}^\Psi(\beta)$ on RHS only — does NOT modify $F(\Psi)$; $G_{eff}/G_N = F(\Psi_{cos})/F(\Psi_{loc})$ unchanged at leading order

Thin-shell criterion:

$\varepsilon \equiv \Delta R/R = (\Psi_\infty - \Psi_{min}) / (6\beta_0 \rho_{int} R^2), \quad \Psi_{min} = \beta_0\rho_{int} / m_{eff}^2$
Light-field regime: $m_{eff} R \approx 3e-05 \ll 1 \Rightarrow \varepsilon \gtrsim 1$, field unscreened, $\Psi \approx \Psi_0$ throughout galaxy



SIM151 — Consistency Summary

Quantity	Result	Status
PRE-CHECK: $\Delta G_{eff}/G_N$ at $\beta_0 = 1.018$	1.56e-08 (1.56e-06%)	PASS $\ll$ 4%
Option B is separate from G_{eff} channel?	YES — force $\propto \beta_0^2$ independent of Λ_0	✓ CONFIRMED
β coupling in Einstein eqns	RHS $T_{\mu\nu}$ only; $F(\Psi)$ unmodified at leading order	✓
Modified KG source	$-\beta(\rho_b)\rho_b$ on RHS — direct baryon sourcing of Ψ	✓
Screening function choice	$f(x) = \tanh^2(x)$: smooth, $f(0)=0$, $f(\infty)=1$, analytically tractable	✓ (Mota & Shaw 2007)
Light-field regime	$m_{eff} \times R_{gal} \approx 3e-05 \ll 1 \rightarrow \varepsilon \gtrsim 1, \Psi \approx \Psi_0$ inside galaxy	✓ CONFIRMED
Fifth-force formula	$a_{total}/a_N = G_{eff}/G_N + 2\beta_0^2 = 1.041 + 2\beta_0^2$	Derived
β_0 required for NGC 3198 ($G_{req} = 3.115$)	$\beta_0^* = \sqrt{(2.074/2)} = 1.0183$	Predicted
β_0 ceiling (Λ_0 constraint)?	NONE — $2\beta_0^2$ channel bypasses Λ_0 ceiling entirely	✓ KEY RESULT
β_∞ bound (Ψ subdominance at z_{drag})	$\beta_\infty \lesssim 2.39e-10$ (binding)	Derived
β_∞ bound ($f_{\sigma 8}$, 1σ Planck+DESI)	$\beta_\infty \lesssim 0.18$ (less binding)	Derived
Parameter space non-degenerate?	$(\beta_0, \beta_\infty, \rho_{sc})$ independently constrained — no degeneracy	✓
Architecture self-consistent?	YES — no algebraic obstruction at derivation level	PASS $\rightarrow$ SIM152/153